Gaussian Mixture Model using Expectation-Maximization (EM) Algorithm

Project Objective

The objective of this project is to implement the Gaussian Mixture Model (GMM) using the Expectation-Maximization (EM) Algorithm from scratch in Base R. The project demonstrates how GMM performs probabilistic clustering by assigning each observation a probability of belonging to each cluster. The implementation is completed without using any external machine learning or clustering libraries.

Software Used

- R
- Base R
- R GUI / RStudio
- Git
- GitHub

Dataset

A synthetic dataset containing **20,000 observations** was generated using Base R.

Features:

- X1
- X2

The dataset contains two Gaussian clusters and is saved as:

synthetic_data.csv

Steps to Run the Program

1. Open **gmm_em_algorithm.R** in R GUI or RStudio.
2. Run the complete program.
3. The synthetic dataset is generated and saved as **synthetic_data.csv**.
4. The EM Algorithm initializes the model parameters.
5. The Gaussian Density Function is calculated.
6. The Expectation (E-Step) computes the responsibility matrix.
7. The Maximization (M-Step) updates the model parameters.
8. The algorithm repeats until convergence or 20 iterations.

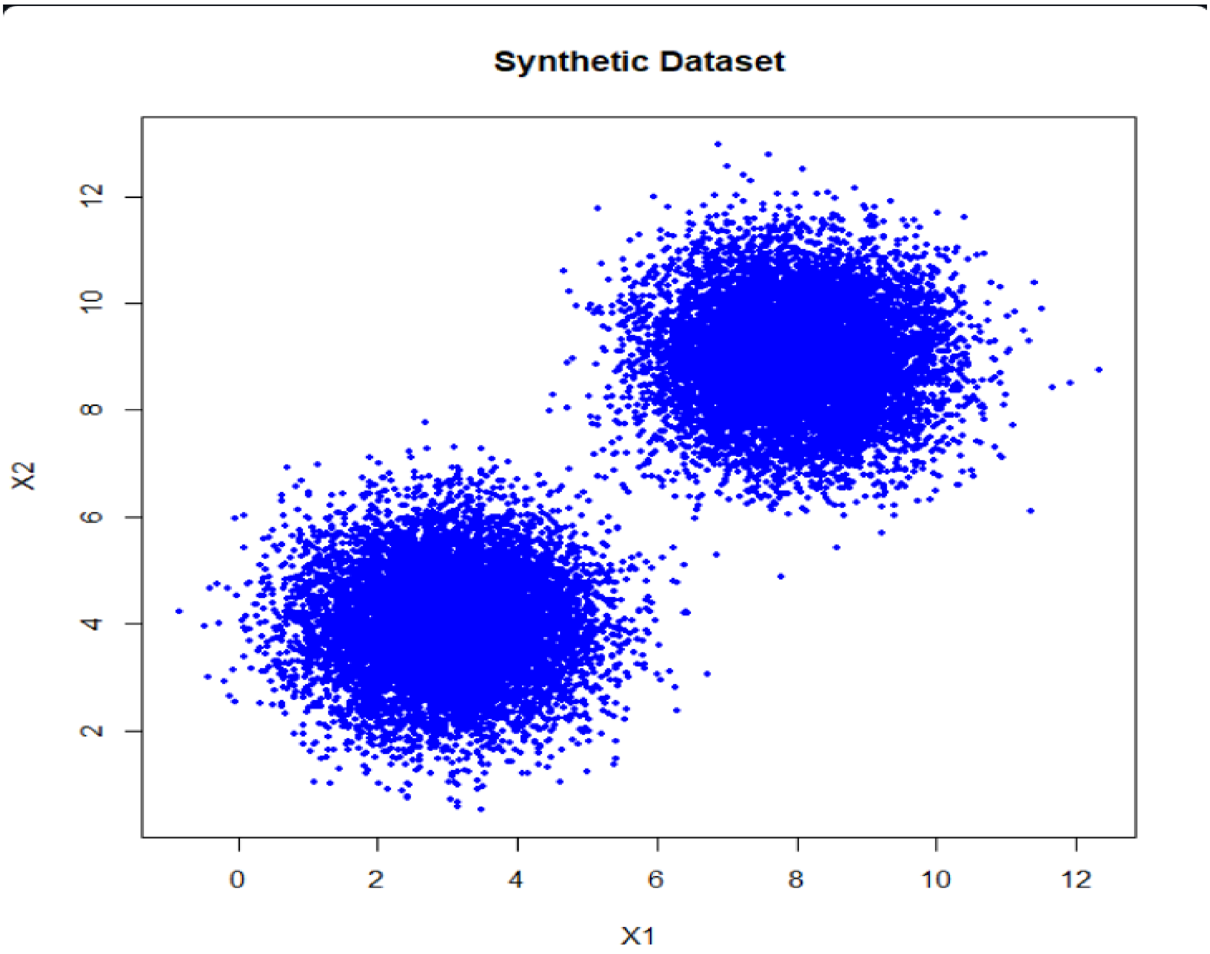
9. The following output plots are generated inside the **Output** folder:
    - dataset.png
    - final_clusters.png
    - log_likelihood.png

Project Folder Structure

```
GMM-EM-Algorithm-Week1/
|
├── README.md
├── gmm_em_algorithm.R
├── synthetic_data.csv
├── Week1_Report.pdf
|
└── Output/
    ├── dataset.png
    ├── final_clusters.png
    └── log_likelihood.png
```

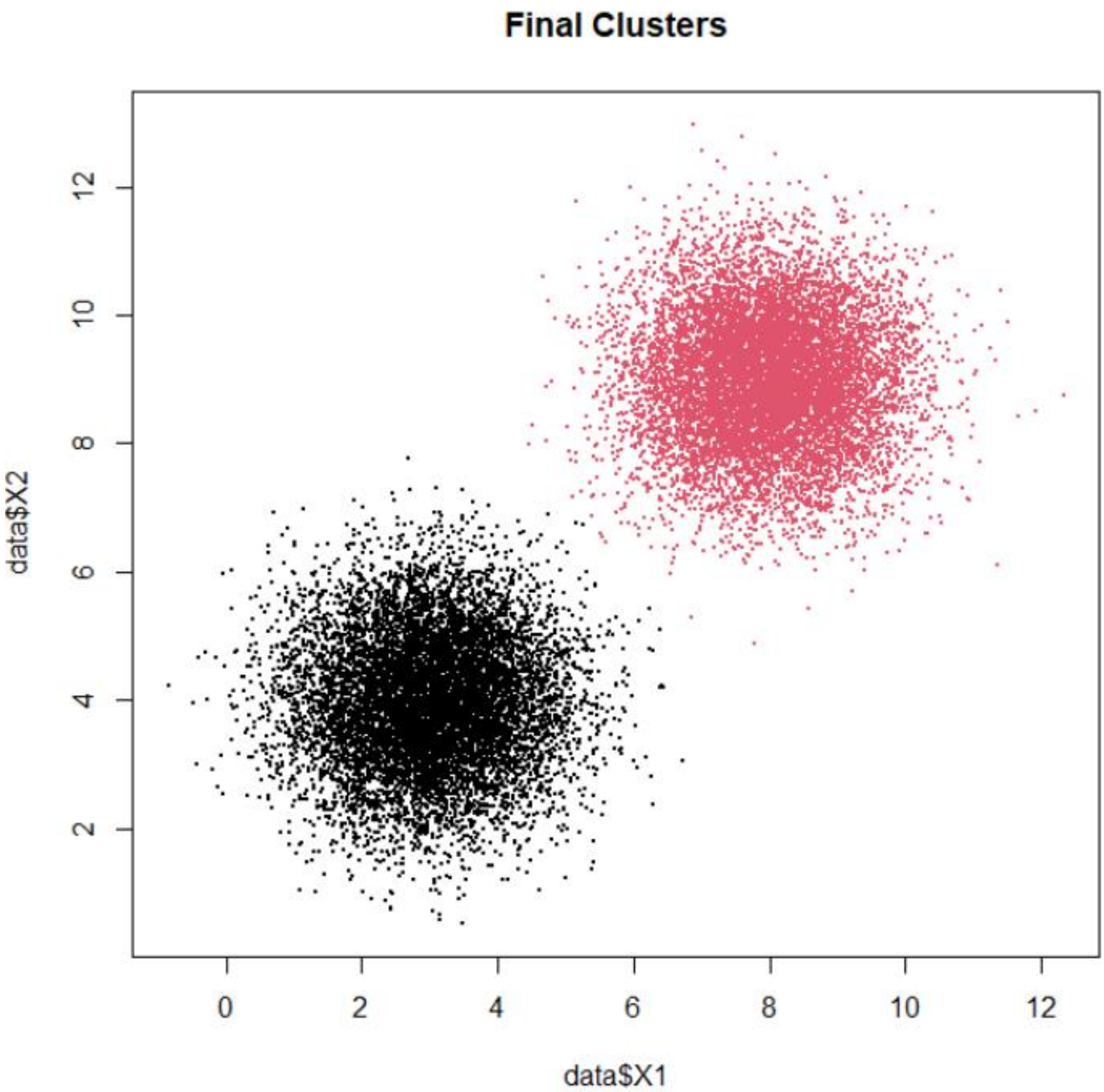
Output Screenshots

1. **Synthetic Dataset**
    - Scatter plot of the generated Gaussian clusters.



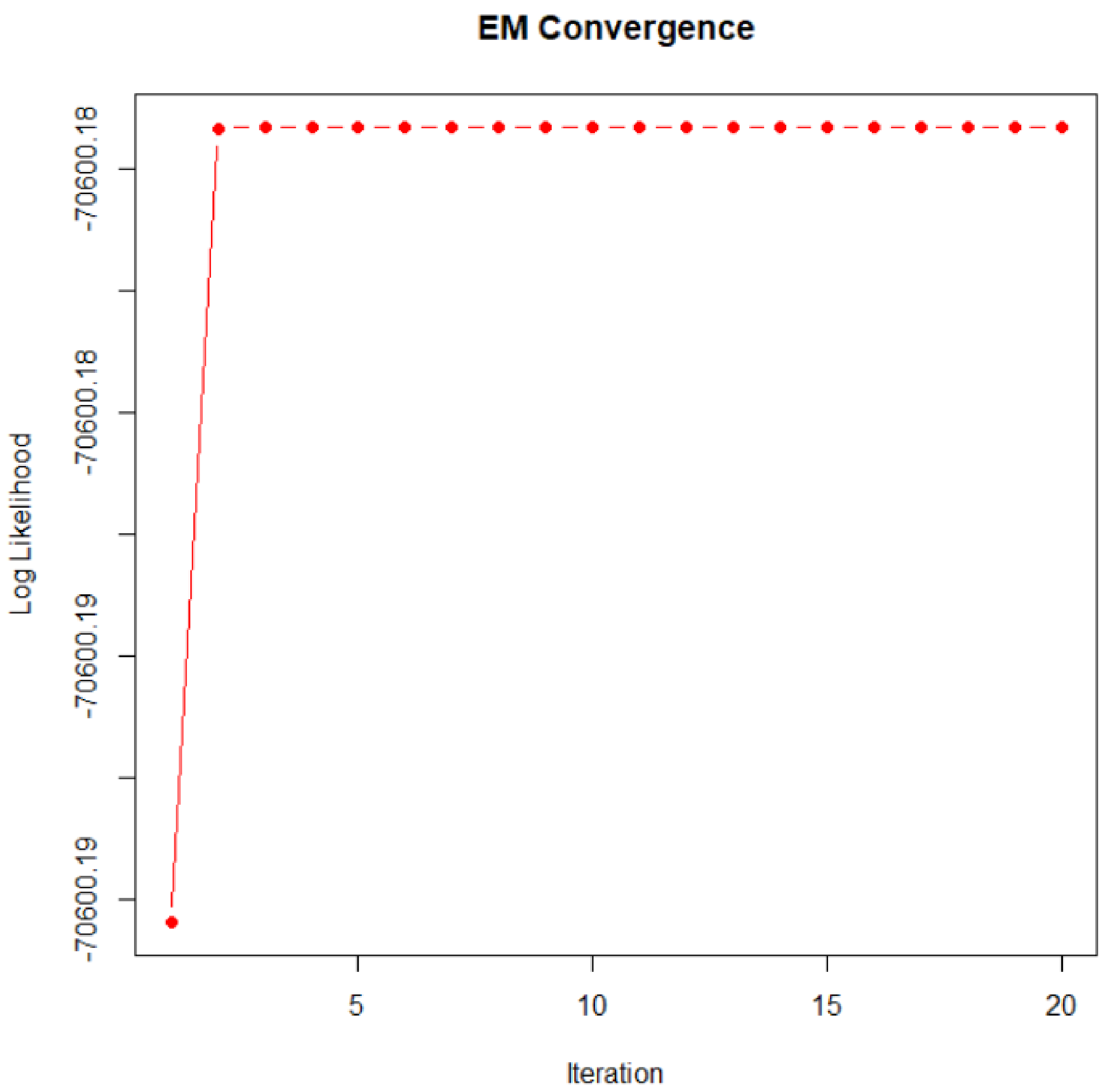
2. Final Cluster Assignment

- Scatter plot showing the clusters identified by the EM Algorithm.



3. Log-Likelihood Plot

- Graph showing the Log-Likelihood value for each iteration until convergence.



Learning Outcomes

After completing this project, I learned to:

- Understand the concept of unsupervised learning.
- Differentiate between K-Means and Gaussian Mixture Models.
- Generate a synthetic dataset using Base R.
- Implement the Multivariate Gaussian Probability Density Function.
- Perform the Expectation (E-Step).
- Perform the Maximization (M-Step).
- Update mixing coefficients, mean vectors, and covariance matrices.
- Compute Log-Likelihood values.

- Check the convergence of the EM Algorithm.
- Visualize clustering results using Base R.
- Upload the completed project to GitHub.

Author

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Course: BCA

Project: Week 1 – Gaussian Mixture Model using EM Algorithm

Language: R (Base R)